United International University (UIU)

Dept. of Computer Science & Engineering (CSE)

Final-Term Exam: Trimester: Fall 2022

Course Code: CSE 4325 Course Title: Microprocessor, Microcontroller and Interfacing
Section: (A, B, C, D, E) Total Marks: 40 Duration: 2 hours

Any examinee found adopting unfair means would be expelled from the trimester/ program as per UIU disciplinary rules.

Question 1: Answer all the questions. (10 Marks)

a.	**Draw** the timing diagram for the memory Read operation of the 8086 microprocessors by showing the activities of Address-Data bus (AD), ALE, $\overline{RD}$, $\overline{DEN}$, $DT/\overline{R}$ in each clock cycle. Also, **explain** the function of the **'ALE'** pin in one to two sentences.	[4+1]
b.	**Explain** the function of the **'READY'** pin in the 8086 microprocessors.	[1]
c.	**Draw** the BIU and EU's cycle for 'MOV <address>' & 'JUMP <address>' instruction in the 8086 microprocessor.	[4]

Question 2: Answer all the questions. (10 Marks)

a. I. **Explain** what changes you have to make in SAP-1 architecture to perform the instruction LDB_{OUT} (Load the content of memory address to B register and show its value in the binary display). Also, **Write** the control words for execution T-states (T4, T5, T6) to perform the instruction [3]

Control Word of a Typical SAP-1 Architecture = $C_pE_p\overline{L_M}\overline{CE}$ $\overline{L_I}\overline{E_I}\overline{L_A}E_A$ $S_UE_U\overline{L_B}\overline{L_O}$

II. **What** are the values of Accumulator registers after executing each instruction programmed in the RAM. [Initially Accumulator = 0H]. [4]

RAM Address	RAM Content		RAM Address	RAM Content
0H	0000 1001		8H	XXXX XXXX
1H	0001 1011		9H	0010 1001
2H	0010 1010		AH	0000 0111
3H	0001 1100		BH	0001 0101
4H	1110 XXXX		CH	0000 0100
5H	1111 XXXX		DH	0011 1100
6H	XXXX XXXX		EH	XXXX XXXX
7H	XXXX XXXX		FH	XXXX XXXX

b. UIU is going to implement a smart entry system. This system has two modules. Student entrance based on their individual ID card is implemented by an outdoor module through RFID sensor. And an indoor module in the program office processes the attendance to calculate total attendance, mark and overall performance. Now, **suggest** a system build for each module whether you will use a microprocessor or microcontroller-based system. Give reasoning using two to three sentences for each build. [3]

Question 3: Answer all the questions. (10 Marks)

a. A microcontroller-based system has two master devices: Master (index:5H) and Master (index:6H). Also, the system has one slave device: Slave (index:11H). Now, in I2C data transfer protocol, Master (index:5H) receives 4-byte data (char 'NotE') from the slave (index:11H). **Draw** [4+1]

	the corresponding sequence diagram. If Master (index:6H) also wants to communicate with the slaves, then **explain** what should happen so that Master (index: 6H) can successfully start communicating with the slave devices.	
b.	I. A microcontroller has a 5V system and any analog value is converted to a 'n' bit digital value. Another microcontroller has a 3.3V system and any analog value is converted to an 8-bit digital value. If the ADC value in the 2nd microcontroller is 151, then **what** will be the ADC report as a value in the 1st microcontroller? II. In the microcontroller, 11 bits digital to analog (DAC) resolution is set. During the digital to analog conversion programming, you have called "analogWrite (1023)" instruction. **Calculate** the duty cycle. **Draw** the duty cycle diagram. Consider the system voltage is 7 V.	[3] [2]
Question 4: Answer all the questions.		**(10 Marks)**
a.	A program having 22 KB in size is written in an advanced processor having a paging unit in its architecture. The OS of the processor saves this program by simply dividing this program into a 6 KB page and assigns each page a virtual page address as 0, 1, 2 and 3. Before execution of this program, the page table in the processor is shown below. While executing this program, the processor first requests virtual page no. 0, then page no. 1, then page no. 2 and lastly page no. 3. **Draw** the final page table after execution of this program. Also, **determine** for which virtual page address, page fault will occur.	[3+2]
b.	An 80286 microprocessor is executing a segment in protected mode and a portion of its local descriptor table is given above. If the current physical address produced by this processor is B247F0H, then **determine** the followings: I. First and last physical address of this segment. II. segment type (CS/SS/DS or ES) and its privilege level.	[3+2]

Virtual Page Address	Real Page Address	
5	0	
-	1	
2	2	Primary memory
-	3	
3	4	
1	5	
7	6	Secondary memory
0	7	
-	8	
4	9	
-	10	

Local Descriptor Table	
Index	**Address**
500H	Base: B56700, Limit: 0129H, Access right: 92H
501H	Base: B23900, Limit: 0F10H, Access right: FFH
502H	Base: B11C00, Limit: 0C12H, Access right: D1H
503H	Base: B67800, Limit: 0560H, Access right: A0H